Alignment package comparison

Arabidopsis SE/PE DAP and bulk ATAC; BWA-MEM2, BWA and Bowtie2

October 16, 2026

Abstract

At the 1-million-pair cap, Bowtie2 gives peak-base Jaccard values of 0.5611 to 0.8493 against BWA-MEM2 across four PE DAP treatments. At the deeper SE DAP cap, FUS3 motif support is 466/998 baseline windows versus 465/998 Bowtie2 windows. These measurements describe agreement and independent motif support, not a validated accuracy ranking. We would retain BWA-MEM2 pending broader evidence.

Setup

Comparisons. BWA-MEM2 2.3, BWA 0.7.19 and Bowtie2 2.5.5 use the same sampled reads and declared reference. The BWA-MEM2 package reports an older banner; recorded binary identity establishes the package version. Bowtie2 uses very-sensitive-local and a 2,000-base maximum PE insert; ATAC also allows dovetail and disables mixed/discordant pairs.

Scoring. Peak Jaccard is base-pair intersection/union against baseline peaks. Fixed-region Spearman compares fragment counts on identical baseline regions. PBM support counts motif-positive 200-base summit windows with same-chromosome, GC-matched controls. These operating policies are not a pure binary substitution.

Table 1. Input identities at the dap-pe-1m cap.

Library	Role / control	Raw templates
SRR31238993	Control	1,000,000
SRR31238589	SRR31238993	1,000,000
SRR31239194	SRR31238993	1,000,000
SRR31238722	SRR31238993	1,000,000
SRR31238919	SRR31238993	1,000,000

1 Bowtie2 changes peak agreement unevenly across treatments

Across 4 PE treatments, fixed-region Spearman spans 0.8793 to 0.9269 for Bowtie2. The 2 lower-peak-count libraries have Jaccard below 0.58, compared with more than 0.82 for the other 2 treatments.

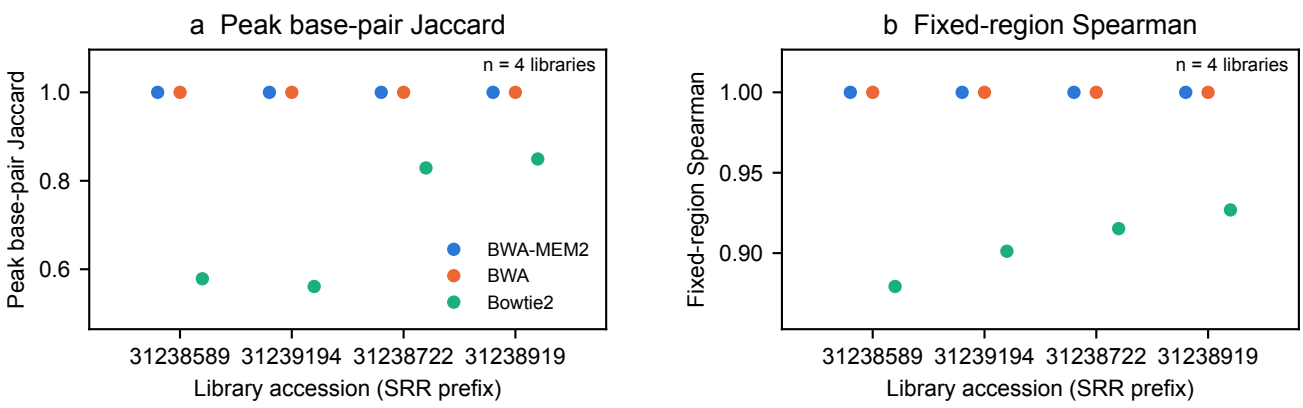


Figure 1. Peak and fixed-region agreement at the 1-million-pair cap.

2 Independent motif support does not favor a general aligner change

The deeper SE DAP comparison covers 2 transcription factors, not 1,998 independent experiments. FUS3 has 2 unmatched control windows, so its top-ranked denominator is 998 rather than 1,000.

Table 2. Independent PBM support in matched top-ranked windows.

Library	Method	Motif hits / windows	Control hits
SRR2926076	bwa	466/998	12
SRR2926076	bowtie2	465/998	13
SRR2926076	baseline	466/998	12
SRR2926841	bwa	581/1000	16
SRR2926841	bowtie2	581/1000	16
SRR2926841	baseline	581/1000	16

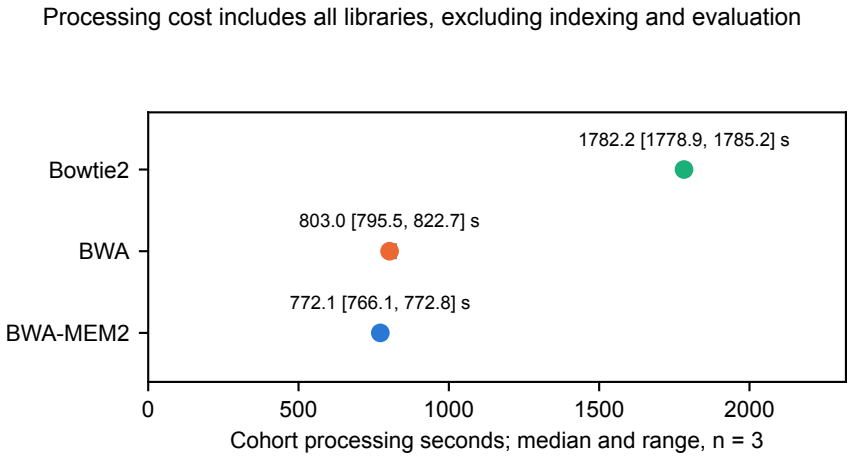


Figure 2. Fresh PE DAP cohort timings at the 1-million-pair cap.

Open issues

- One species and selected TFs cannot establish a universal aligner ranking.
- Bowtie2 local alignment and pairing policies differ from BWA policies.
- Timings cover all cohort libraries, exclude indexing and evaluation, and allow 14 tool CPUs.

Recommendation

We would retain BWA-MEM2 and investigate disagreement on held-out biological evidence.

1. Expand TF and read-length coverage.
2. Inspect alternative-specific peaks before accepting greater retention as improvement.

Appendix: Methods

We summarize recorded results without rerunning biological processing. The main Arabidopsis matrix uses seeded nested raw-template subsets (seed 60141), three fresh technical repetitions at the 1-million cap and one at the 5-million cap. Actual counts replace requested caps for short libraries. DAP retains primary mapped reads including MAPQ 0; ATAC selects proper primary pairs with both mates MAPQ at least 30, excludes MAPQ 255, organelles and marked duplicates. Biological identity is preserved. Source files are SHA-256 checked against the accompanying catalog; evidence snapshot revision ed1c50b8e4b5 includes the October 13 to 16 records. Completed local reference assessments are separately identified in that catalog.